

Sprint 2 Readiness

Project: Image Guided Surgery Intervention

Sprint 2 Goals

- Stylus Tracking: Enable real-time 3D stylus tracking and AR overlay by completing the Camera-to-Stylus calibration pipeline.
- Hardware Integration & Testing: Validate the software pipeline using the physical ultrasound scanner and positional NDI tracker.
- 3D Volume Reconstruction: Develop the 3D volume reconstruction pipeline by integrating PlusToolkit for real-time data acquisition and VTK for volumetric rendering.
- Architecture Maintenance: Perform continuous code refactoring and generate comprehensive documentation for new modules to ensure extensibility.

Backlog Review

This table reviews the status of Sprint 1 work and introduces the new work planned for Sprint 2.

Story ID	Title	Status in Sprint 1	Actions for Sprint 2	Notes/Reason
S1-01	Familiarization with Existing Software	Completed	None	Team has reviewed existing repo and documentation.
S1-02	Setup and Documentation	Completed	None	Development environments are reproducible and setup procedure is documented.
S1-03	Modularization and Extensibility	Completed	None	Core components such as OverlayApp.py and OverlayRenderer.py, have been refactored, further refactoring moved to new continuous maintenance story.
S1-04	Database Integration	Completed	None	Database used to store patient scans integrated.
S2-11	Stylus Pivot & Axial Calibration	Not started	Validate Calibration	Finalize the transformation matrix calculation and verify the tip offset error is negligible.
S2-12	Visual AR Validation Overlay	Not Started	Implement VTK Overlay	Develop the rendering loop to display 3D shapes over the tracked ArUco stylus tip.
S2-21	XML Configuration Management	Done	Maintain/Up date	Ensure the ultrasound wand + Aurora NDI tracker XML is configured for hardware port mapping.

S2-22	Data Capture	In Progress	Refine Start/Stop UI	Use PlusToolKit server that accepts wand + tracker configuration file, and NDI Aurora tracker + wand output data stream, to produce an mha tracked ultrasound sequence metadata file to be reconstructed.
S2-23	Calibrate/Align Data Capture	Not started	Automate Calibration	Perform temporal calibration on sequence metadata file to ensure the tracking data and ultrasound frames are aligned by timestamp.
S2-24	Capture Session Data	Not started	Add new tables to database	Ensure database is configured to store session data from tracked sequence captures.
S2-25	Automated Volume Generation	Not Started	Bash Script Scripting	Write and test the bash script to automate PlusToolkit's VolumeReconstructor execution.
S2-26	Volumetric Visualization	Not Started	VTK GUI Integration	Integrate the vtkVolume rendering pipeline into the main application dashboard, and implement transfer

Sprint Planning Details

Breakdown of tasks selected for Sprint 2, assigned based on the Roles/RACI chart in the Team Charter.

Story/Task	Assigned To	Est. Points	Status
S2-11: Stylus Pivot & Axial Calibration	Muneeb Asif	5	Ready
S2-12: Visual AR Validation Overlay	Muneeb Asif	8	Ready
S2-21: XML Configuration Management	Hannah Taylor	3	Ready
S2-22: Data Capture	Janki Patel	3	Ready
S2-23: Calibrate Data Capture	Delaney McLachlan	3	Ready
S2-24: Automated Volume Generation	Delaney McLachlan	5	Ready
S2-25: Volumetric Visualization	Elise Drouillard	8	Ready

Sprint Documentation Assignment

Breakdown of the assigned documentation work for sprint 2.

Delaney McLachlan: High Level Architecture Diagram, Sprint 2 Feature Document

Janki Patel: Updating Backlog, Product Increment Template

Hannah Taylor: High Level Object Design, Burn Down Chart

Muneeb Asif: Sprint Review and Retrospective

Elise Drouillard: Update Sprint 2 Readiness Document, Feedback Review Document